UTM
UNIVERSITI TEKNOLOGI MALAYSIA

COURSE INFORMATION

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School/Faculty:	**Faculty Computing**		
Program name:	**Bachelor in Software Engineering with Honours**		
Course code:	**SCSE2123**	**Academic Session/Semester:**	**2025/2026/1**
Course name:	**Software Requirements Engineering**	**Pre/co requisite (course name and code, if applicable):**	
Credit hours:	**3**		

Course synopsis	This course introduces students to the fundamental principles and practices of software requirements engineering within the software development lifecycle. It builds on basic programming and software design knowledge, equipping learners with skills to handle real-world software projects effectively. The course explores requirements elicitation techniques, including stakeholder interviews, surveys, and workshops to gather user needs. It covers analysis and modeling of functional and non-functional requirements using tools like UML diagrams, use cases, and user stories. Additional topics include requirements specification writing, validation and verification methods, risk assessment, and requirements management in both traditional and agile environments. The curriculum also addresses prototyping, traceability, and handling changes to ensure requirements align with project goals. The course is delivered through a combination of lectures, interactive tutorials, hands-on labs, and team-based projects, with an emphasis on collaborative problem-solving and practical application of requirements engineering tools. Upon completion, students will be able to elicit, analyze, and document software requirements using industry-standard techniques and evaluate their quality for completeness, consistency, and feasibility. They will also design and manage requirement specifications that support effective software development and testing.
Course coordinator (if applicable)	Dr. Noraini Ibrahim

Course lecturer(s)	**Name**	**Office**	**Contact no.**	**E-mail**
	Dr Noraini Ibrahim	N28, FC		noraini_ib@utm.my
	Associate Professor Dr. Radziah Mohamad	N28a, FC		radziahm@utm.my

Mapping of the Course Learning Outcomes (CLO) to the Programme Learning Outcomes (PLO), Teaching & Learning (T&L) methods and Assessment methods:

No.	CLO*	PLO (MQF Cluster Code)	Taxonomies and Graduate attributes	T&L methods	Assessment methods
CLO1	Analyze the different requirement types for various application domains during requirements engineering process.	PLO2 (C2)	C4	Active learning, Design thinking	Problem Solving (5%), Mid-Term Test (15%)
CLO2	Create an innovative solution for the real-world problem using appropriate requirements engineering method.	PLO3 (C3A)	C6 P5	Design thinking, Project - case study, Work-based learning	PR1 URD (10%) PR2 Req. model (10%) PR3 SRS (12%) AA High Fidelity Prototype (30%)
CLO3	Ability to lead and work effectively in a team, and to present orally the requirements engineering deliverables	PLO8 (C3F)	A(CC2)	Collaborative team discussion, Self & Peers reflection	3 x Leadership assessment : 10%

PR – Project; AA – Alternative Assessment.

MQF Cluster Code
C1 = Knowledge & Understanding, **C2** = Cognitive Skills, **C3A** = Practical Skills, **C3B** = Interpersonal Skills, **C3C**= Communication Skills, **C3D** = Digital Skills, **C3E** = Numercy Skills, **C3F** = Leadership, Autonomy & Responsibility, **C4A** = Personal Skills, **C4B** = Enterpreneurial Skills, **C5** = Ethics & Professionalism

Details on Innovative T&L practices:

No.	Type	Implementation
1	Active Learning	Conducted activity to engage students in a new topic, activate prior knowledge, share unit objectives, and monitor students' learning.
2	Design thinking	Conducted collaborative group activity where students effectively teach each other (with the teacher's guidance) a given skill or procedure, topic or problem.
3	Project - case study	Conducted based on given real world problem. The students must manage their projects by submitting specified deliverables based on the given problem. The problems are given in sequential steps throughout the semester.
4	Work-based learning	Involvement from real stakeholders from industry during requirements elicitation, analysis, and validation
5	Research-based	Learners engaged through research activities based on a Software Engineering Project-based, Problem-based Learning (PoPbL) pedagogical approach with Design Thinking (DT) activities. Analysis of stakeholder profiles for project case study scenario on the existing system and it's context in the stakeholder business environment is conducted. Comprehensive requirements model document by representing the elicited requirements is delivered.
6	Collaborative team discussion	Learners use an online platform to interact with peers by posting thoughts or comments in response to their project discussion.
7	Self & Peers reflection	Learners assessing their own work and provide feedback to each other on their work.

Transferable skills (generic skills learning in course of study which can be useful and utilised in other settings) :
Leadership skills

Student learning time (SLT) / Effective Learning Time (ELT) details:

Week/ Meeting	Course Content Outline and Subtopics	CLO	Learning and Teaching Activities										TOTAL SLT
			Face-to-Face (F2F)								Non F2F Independent Learning		
			Physical				Online (Synchronous)				Online (Asynchronous)	Others	
			L	T	P	O	L	T	P	O			
Week 1	**Topic 1: Introduction and Overview** ● Requirements Engineering: What, Why, Where, How ● The Role and Tasks of a Requirements Engineer *** Assessment: PS1 (2.5%)**	CLO1	3								2	2	7
Week 2	**Topic 2: Fundamental Principles** ● Overview of principles ● Stakeholders ● System and Context ● Sustainability Analysis for System Domains - Environmental, social, and economic contexts *** Assessment: PS2 (2.5%)**	CLO1	3								2	2	7

Week 3	**Topic 3: Practices for Requirements Elicitation** ● Sources for requirements – stakeholders, documents, other systems ● Requirements categorization using Kano Model	CLO2	3								2	2	7
Week 4	**Topic 3: Practices for Requirements Elicitation (continue)** ● Gathering Techniques ● Design and Idea-Generating Techniques	CLO2	3								2	2	7
Week 5	**Topic 3: Practices for Requirements Elicitation (continue)** ● Elicitation workshop by involved stakeholders ● Problems description by stakeholders *** WBL#1: Stakeholder elicitation workshop and brainstorming**	CLO2	3								2	2	7
Week 6	**Topic 4: RE Process Facets** ● Influencing Factors ● Requirements Engineering Process Facets (Time, Purpose, Target) **Topic 5: Work Products in RE** ● Work Products in Requirements Engineering	CLO2	3								2	3	8
Week 7	**Topic 6: Practices for Model-Based Requirement Documentation (Part 1)** ● Definition of model, benefits, roles in RE ● Development of Goal model ● Development of Business Process Model (Swimlane diagram) *** Assessment: PR1 - (User Requirements Document – URD 10%) &**	CLO2, CLO3	3								2	2	7

Week 8	**MID SEM BREAK**												
Week 9	**Topic 6: Practices for Model-Based Requirement Documentation (Part 2)** ● Introduction to Model-Based Work Products ● Modeling System Context - Use Case diagram, Use case specification (UCS), User Story (revision from SEP) ● Modeling Function and Flow –Activity diagram *** Assessment: PR2 - (Requirements Model Document 12%) &**	CLO2, CLO3	3							2	2	7
Week 10	**Topic 6: Practices for Model-Based Requirement Documentation (Part 2-continue)** ● Modeling Structure and Data – Domain class diagram ● Modeling Behaviour - Sequence diagram	CLO2, CLO3	3							2	2	7
Week 11	**Topic 7: Practices for Text-Based Requirement Documentation (Part 1)** ● Natural-Language-Based Work Products ● Template-Based Work Products - Phrase Templates, Form Templates, Document Templates ● Advantages and Disadvantages	CLO1, CLO2	3							2	2	7
Week 12	**Topic 7: Practices for Text-Based Requirement Documentation (Part 2)** ● Requirements Documents and Documentation Structures ● Prototypes in Requirements Engineering ● Quality Criteria for Work Products and Requirements *** Assessment: PR3- (Software Requirements Specification-SRS 10%) &**	CLO2, CLO3	3							1	2	6

Week 13	**Topic 8: Practices for Requirements Validation and Negotiation** ● Fundamentals of Requirements Negotiation ● How Do You Resolve a Requirements Conflict? ● Conflict Types ● Conflict Resolution Techniques ● Fundamentals of Requirements Validation ● Important Aspects for Validation ● Validation Techniques *** Assessment Briefing: Alternative Assessment (High Fidelity Mock-up**	CLO1	3							2	2	7
Week 14	**Topic 9: Practices for Requirements Management** ● What is Requirements Management? ● Life Cycle Management ● Version Control ● Configurations and Baselines ● Attributes and Views ● Traceability ● Handling Change ● Requirements Management Tools (with Generative AI features) *** WBL#3: Stakeholder**	CLO1	3							2	3	8
Week 15	*** Alternative Assessment Submission (High Fidelity Mock-up Prototype 40%)**	CLO2										0
										SUB-TOTAL SLT :		**92**

Continous Assessment		%	Face-to-Face (F2F)		NF2F Independent Learning for Assessment		TOTAL SLT
			Physical	Online (Synchronous)	Online (Asynchronous)	Others	
1	Problem Solving 1 (PS1)	2.5	0.5				0.5
2	Problem Solving 2 (PS2)	2.5	0.5				0.5
3	PR1 User Requirements Document (URD)	12				2	3
4	PR2 Requirements Model Document	13				3	3
5	PR3 Software Requirements Specification (SRS) Document	15				3	3
6	Mid Term Test	15	2			4	6
					SUB-TOTAL SLT :		**16**

Summative Assessment		%	Face-to-Face (F2F)		NF2F Independent Learning for Assessment		TOTAL SLT
			Physical	Online (Synchronous)	Online (Asynchronous)	Others	
1	Alternative Assessment High Fidelity Mock-up	40	2	5		5	12
					SUB-TOTAL SLT :		**12**
					SLT for Assessment:		**28**
					GRAND TOTAL SLT:		**120**

A	% SLT for F2F Physical Component	36.97
B	% SLT for Online & Independent Learning Component :	63.03
C	%SLT for Online Component:	25.00
D	% SLT for All Practical Component:	0.00
D1	% SLT for F2F Physical Practical Component:	0.00
D2	% SLT for F2F Online Practical Component:	0.00
Please tick (/) if this course is Industrial Training/ Clinical Placement/ Practicum using 50% of Effective Learning Time (ELT)		

Identify special requirement or resources to deliver the course (e.g.,software,nursery, computer lab, simulation room etc)

Software: Enterprise Architect, Cloud Application: Github, Modern Requirement (with Gen.AI)

References (include required and further readings, and should be the most current)

Main references

1. Certificate Professional Requirements Engineering (CPRE) Foundation Level Handbook by International Requirements Engineering Board (IREB). https://cpre.ireb.org/en/knowledge-and-resources/downloads#cpre-foundation-level-handbook (2024).

Additional references

2. Phillip A. Laplante and Kassab.M, Requirements Engineering for Software and Systems, 4th Edition. New York, USA: Auerbach Publications, 2022.
3. Fred Heath, Managing Software Requirements the Agile Way, Packt Publishing, 2020

K. Wiegers and J. Beatty, Software Requirements, 3rd Edition. Washington: Microsoft Press, 2014.

4. H. Gomaa, Software Modeling and Design: UML, Use Cases, Patterns, and Software Architectures. New York, USA: Cambridge University Press, 2011.

Learning Management System (LMS)

http://elearning.utm.my

Other additional information (if applicable)
Academic honesty and plagiarism: *(Below is just a sample)*
Assignments are individual tasks and NOT group activities (UNLESS EXPLICITLY INDICATED AS GROUP ACTIVITIES) Copying of work (texts, simulation results etc.) from other students/groups or from other sources is not allowed. Brief quotations are allowed and then only if indicated as such. Existing texts should be reformulated with your own words used to explain what you have read. It is not acceptable to retype existing texts and just acknowledge the source as a reference. Be warned: students who submit copied work will obtain a mark of zero for the assignment and disciplinary steps may be taken by the Faculty. It is also unacceptable to do somebody else's work, to lend your work to them or to make your work available to them to copy.

Other additional information (if applicable)

1. To pass this course, students must pass both Continuous and Final assessment.

2. All students must adhere to the following to pass a computing course:
i. Pass both continuous assessment AND final assessment
ii. Get at least minimum pass marks following UTM regulation (40%)
- The passing mark for Continuous Assessment is 30% of continuous marks.
- The passing mark for the Final Assessment is 20% of the final marks.-
- If a student does not achieve the passing mark for continuous assessment or the passing mark for the final assessment, then the student's overall final mark will be set below the UTM passing mark.
Attendance is compulsory and will be taken in every lecture session. Student with less than 80% of total attendance is not allowed to sit for final exam.

3. Students are required to behave and follow the University's dressing regulation and etiquette all the time.

4. Exercises and tutorial will be given in class and some may be taken for assessment. Students who do not do the exercise will lose the coursework marks for the exercise.

5. Assignments must be submitted on the due dates. Some points will be deducted for late submissions. Assignments submitted three days after the due date will not be accepted.

6. Make up exam will not be given, except to students who are sick and submit medical certificate confirmed by UTM panel doctors. Make up exam can only be given within one week of the initial date of exam.

Disclaimer:

All teaching and learning materials associated with this course are for personal use only. The materials are intended for educational purposes only. Reproduction of the materials in any form for any purposes other than what it is intended for is prohibited.
While every effort has been made to ensure the accuracy of the information supplied herein, Universiti Teknologi Malaysia cannot be held responsible for any errors or omissions.

Prepared by:	**Certified by:**
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Signature: Noraini	**Signature:**
Date: 06-Oct-25	**Date:** 6/10/2025